SECURITY INCIDENT REPORT

C2 Beaconing Detection — Home Lab Investigation

Prepared by: Sanjeev Kushwaha | Date: 27 May 2026 | Classification: Lab / Training

1. Incident Summary

ALERT — Wazuh Rule 92213 | Severity Level 15 (Critical)
PowerShell initiating outbound C2 beaconing detected from endpoint DESKTOP-Q6GKP4D. MITRE ATT&CK technique T1105 — Ingress Tool Transfer / Command and Control confirmed. Alert fired 101 times. Immediate investigation conducted.

Field	Detail
Incident ID	INC-2026-0527-001
Date & Time (UTC)	2026-05-27 05:58:38
Analyst	Windsan
Affected Endpoint	DESKTOP-Q6GKP4D
Victim IP	192.168.211.129
Attacker / C2 IP	192.168.211.132 (Kali Linux)
C2 Port	4444 (TCP)
Detection Source	Wazuh SIEM + Sysmon + Wireshark
Severity	Critical — Level 15
Status	Investigated — Contained (Lab Environment)

2. What is C2 Beaconing?

Command and Control (C2) beaconing is a technique used by attackers after compromising a machine. The infected machine makes periodic, automated outbound network connections to an attacker-controlled server at regular intervals — like a heartbeat. This allows the attacker to maintain persistent access, issue commands, and exfiltrate data over time.

The key characteristic that makes beaconing detectable is its regularity. Unlike human-generated traffic which is irregular and unpredictable, malware beacons at machine-precision intervals — every 30 seconds, every 5 minutes — even at 3am when no user is active. This pattern is visible in packet captures and is one of the most important signals an analyst learns to identify.

MITRE ATT&CK maps this behaviour under:

- Tactic: Command and Control (TA0011)
- Technique: Ingress Tool Transfer (T1105)
- Technique: Application Layer Protocol (T1071)

---

3. Lab Environment

Component	Role	IP Address	Platform
Kali Linux	Attacker / C2 Server	192.168.211.132	VMware
Windows 10	Victim / Compromised Endpoint	192.168.211.129	VMware
Wazuh	SIEM / XDR	Wazuh Server	VirtualBox
Sysmon	Endpoint Telemetry	Installed on Windows 10	VMware
Wireshark	Network Capture	Running on Kali	VMware
Windows 11	Physical Host	Host Machine	Bare Metal

4. Attack Simulation

4.1 C2 Server Setup (Kali)

A netcat listener was established on Kali Linux to act as the C2 server, accepting inbound beacon connections from the victim machine:

```
while true; do nc -lvp 4444; done
```

The listener was configured in a loop to accept repeated connections — simulating a persistent C2 server that remains available across multiple beacon cycles.

4.2 Beacon Script (Windows 10 Victim)

A batch script was created at C:\Users\Public\beacon.bat on the victim machine. The script simulated malware behaviour by making outbound TCP connections to the C2 server every 30 seconds:

```
@echo off :loop powershell -Command "Test-NetConnection -ComputerName 192.168.211.132
-Port 4444" timeout /t 30 /nobreak goto loop
```

The script used PowerShell's Test-NetConnection cmdlet to initiate TCP connections — this is significant because PowerShell initiating outbound connections to non-standard ports is a high-fidelity indicator of compromise in real environments.

5. Evidence & Findings

5.1 Network Evidence — Wireshark

Wireshark was used to capture and analyse all traffic between the victim and the C2 server. The capture filter applied was: tcp.port == 4444 combined with ip.addr == 192.168.211.129.

Wireshark Field	Observed Value	Significance
Source IP	192.168.211.129	Victim machine — DESKTOP-Q6GKP4D
Destination IP	192.168.211.132	C2 server — Kali Linux
Destination Port	4444	Non-standard port — red flag
Protocol	TCP	Reliable connection-based comms
TCP Flags	SYN, SYN-ACK, FIN	Full beacon cycle per interval
Payload	0 bytes (empty)	Check-in only — no data yet
Source Port	50260	Matches Sysmon log — corroborated
Interval	~30 seconds	Machine-precision — automated
MAC Vendor	VMware	Internal lab network confirmed

The TCP stream follow confirmed 0 client packets, 0 server packets — the beacon was a pure connection check-in with no data exchanged. This is consistent with an implant awaiting commands from the operator.

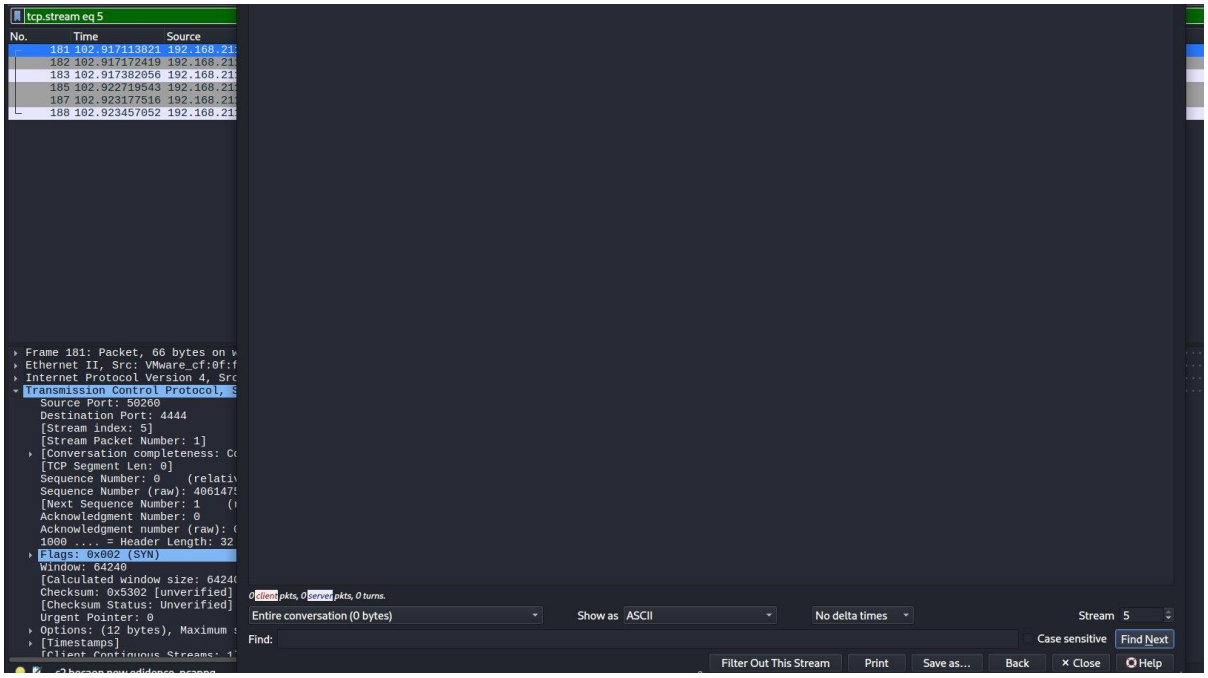


Figure 1 — Wireshark TCP stream showing SYN beacon to port 4444, 0 bytes payload, source port 50260 confirmed

5.2 Endpoint Evidence — Sysmon Event ID 3

Sysmon was queried on the Windows 10 victim machine via Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational. Filtering for Event ID 3 (Network Connection) revealed the following critical log entry:

Sysmon Field	Value
Event ID	3 — Network Connection
Timestamp (UTC)	2026-05-27 05:58:38.453
Process	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Process ID	1420
User	DESKTOP-Q6GKP4D\windsan
Source IP	192.168.211.129
Source Port	50260
Destination IP	192.168.211.132
Destination Port	4444
Protocol	TCP
Initiated	True (outbound)
Computer	DESKTOP-Q6GKP4D
Event Record ID	9374

Key finding: Source Port 50260 in Sysmon matches Source Port 50260 in Wireshark — two independent data sources corroborating the same connection event. This cross-source correlation is how analysts confirm evidence with high confidence.

5.3 SIEM Evidence — Wazuh Alerts

The Wazuh SIEM dashboard was queried for alerts from agent DESKTOP-Q6GKP4D. Two significant findings were identified:

Rule ID	Description	Level	Times Fired	MITRE ID	Verdict
92213	Executable file dropped in folder commonly used by malware	15 — Critical	101	T1105	TRUE POSITIVE
92217	Executable dropped in Windows root folder	6 — Medium	51+	T1570	FALSE POSITIVE

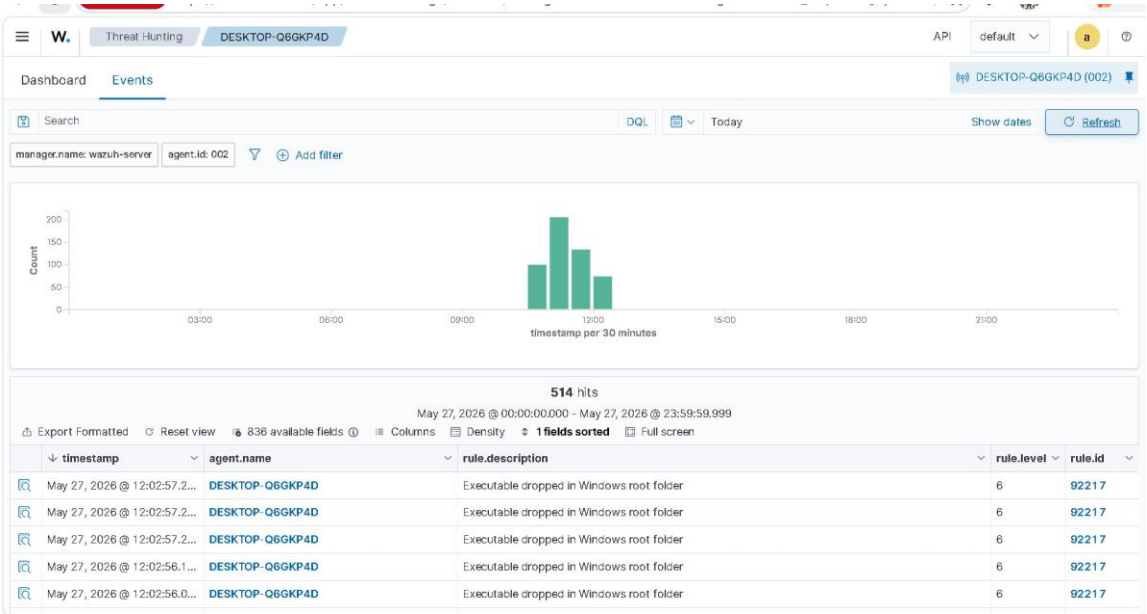


Figure 2 — Wazuh Threat Hunting dashboard showing 514 events from DESKTOP-Q6GKP4D, Rule 92213 Level 15 firing repeatedly at 12:02

Rule 92213 (Level 15) fired because PowerShell dropped a temporary .ps1 script policy test file into AppData\Local\Temp — a directory heavily associated with malware staging. Wazuh automatically mapped this to MITRE T1105 (Ingress Tool Transfer) and flagged it for email notification (rule.mail: true), which in a production SOC would have paged the on-call analyst immediately.

Rule 92217 was determined to be a false positive — mscorsvw.exe (.NET optimization service) creating DLL files in the Windows assembly cache. This is legitimate Windows behaviour. Triaged and closed as false positive.

6. Attack Timeline

Time (UTC)	Event	Source
05:58:36	PowerShell drops __PSScriptPolicyTest.ps1 into AppData\Temp	Sysmon Event ID 11
05:58:38	PowerShell initiates TCP connection to 192.168.211.132:4444	Sysmon Event ID 3
05:58:38	SYN packet observed in Wireshark — beacon cycle 1	Wireshark
05:58:38	Wazuh Rule 92213 fires — Level 15 Critical	Wazuh SIEM
06:08:39	Beacon repeating every 30 seconds — pattern confirmed	Wireshark
06:32:56	Wazuh Rule 92217 fires — false positive triaged	Wazuh SIEM
Ongoing	Beacon loop continues — 101 alerts generated total	Wazuh SIEM

---

7. MITRE ATT&CK Mapping

Tactic	Technique ID	Technique Name	Evidence
Command & Control	T1105	Ingress Tool Transfer	Wazuh Rule 92213, PowerShell .ps1 drop
Command & Control	T1071	Application Layer Protocol	TCP beaconing via port 4444
Lateral Movement	T1570	Lateral Tool Transfer	Wazuh Rule 92217 (false positive — triaged)
Execution	T1059.001	PowerShell	powershell.exe initiating all connections

8. Detection Gap Identified

DETECTION GAP — No Wazuh Rule for Outbound PowerShell on Non-Standard Ports
Wazuh received Sysmon Event ID 3 logs showing PowerShell connecting to port 4444. However no existing rule matched this pattern and no alert was generated from the network connection event alone. The attack was only caught indirectly via the file creation event (Event ID 11). This represents a gap in detection coverage.

Recommendation: Create a custom Wazuh rule to alert on PowerShell.exe initiating outbound TCP connections to ports outside 80, 443, and 8080. Suggested rule logic:

- EventID = 3 (Sysmon network connection)
- Image contains powershell.exe
- DestinationPort NOT IN (80, 443, 8080, 53)
- Initiated = true (outbound only)
- Suggested severity: Level 12 — High

---

9. Analyst Verdict & Recommendations

9.1 Verdict

Finding	Verdict
C2 beaconing activity	CONFIRMED — True Positive
Rule 92213 alert	CONFIRMED — True Positive
Rule 92217 alert	FALSE POSITIVE — Legitimate Windows process
Detection gap on Event ID 3	CONFIRMED — Custom rule recommended

9.2 Recommendations

- Write a custom Wazuh detection rule for PowerShell outbound connections to non-standard ports
- Add port 4444 to network monitoring watchlist — commonly used by Metasploit and other C2 frameworks
- Implement PowerShell script block logging (Event ID 4104) on all endpoints for deeper visibility
- Consider application whitelisting to prevent unauthorised .bat and .ps1 script execution
- Tune Rule 92217 to exclude known legitimate processes like mscorsvw.exe to reduce false positive noise
- In a production environment this activity would warrant immediate endpoint isolation and escalation to L2

10. Evidence Artefacts

Artefact	Description	Location
c2_beacon_evidence.pcap	Full packet capture of beaconing traffic	Kali Desktop
Wireshark Screenshot	TCP stream — port 4444, SYN flags, 0 bytes	Figure 1 — this report
Sysmon Event ID 3	PowerShell network connection log	DESKTOP-Q6GKP4D Event Viewer
Sysmon Event ID 11	PowerShell file creation in Temp	DESKTOP-Q6GKP4D Event Viewer
Wazuh Rule 92213	Level 15 SIEM alert — T1105 C2	Figure 2 — this report
beacon.bat	Simulated malware beacon script	C:\Users\Public\beacon.bat

— End of Report —

Analyst: Windsan | Lab: Home SOC | Date: 27 May 2026